

Crime Rate Prediction and Safety Navigation Using Machine Learning and Google Map

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ABSTRACT

This project focuses on Crime Rate Prediction and Safety Navigation using Google Maps, aiming to enhance public safety in urban environments. The system integrates machine learning techniques with spatial data from the Google Maps API to identify crime-prone regions and suggest safer travel routes. Historical crime records, combined with environmental and traffic-related features such as population density, road conditions, street lighting, and accident hotspots, are used to train predictive models including Support Vector Machine (SVM), Random Forest, and Neural Networks. Each area is classified into different crime risk levels, which are then visualized through heatmaps and integrated into Google Maps routing services. Instead of relying solely on the shortest or fastest route, the system provides users with alternative navigation paths that minimize exposure to high-crime areas. Experimental results demonstrate that the approach successfully reduces the probability of traveling through unsafe regions while maintaining reasonable travel times. This work highlights the potential of combining crime analytics with navigation systems to support crime prevention strategies and empower individuals with safer mobility choices. Future enhancements include incorporating real-time data, applying advanced deep learning models, and expanding the system for deployment across multiple cities.

Keywords: Crime prediction, Machine Learning, Random Forest, SVM, Decision Tree, Crime analytics, Predictive policing.

I. INTRODUCTION

Crime is a significant concern affecting community safety, economic development, and urban planning. Predicting crime patterns enables authorities to deploy resources more efficiently and take preventive measures (Fig. 1). Traditional crime analysis techniques depend heavily on manual investigation, which is time consuming and limited in scope. Machine Learning (ML) offers the capability to analyze complex, multidimensional datasets and capture hidden crime patterns related to time, locality, population density, and socio-economic indicators [1]. This research focuses on developing a crime prediction model using only classical ML algorithms (excluding boosting methods), with Random Forest showing superior performance due to its ensemble nature and ability to handle heterogeneous crime data.

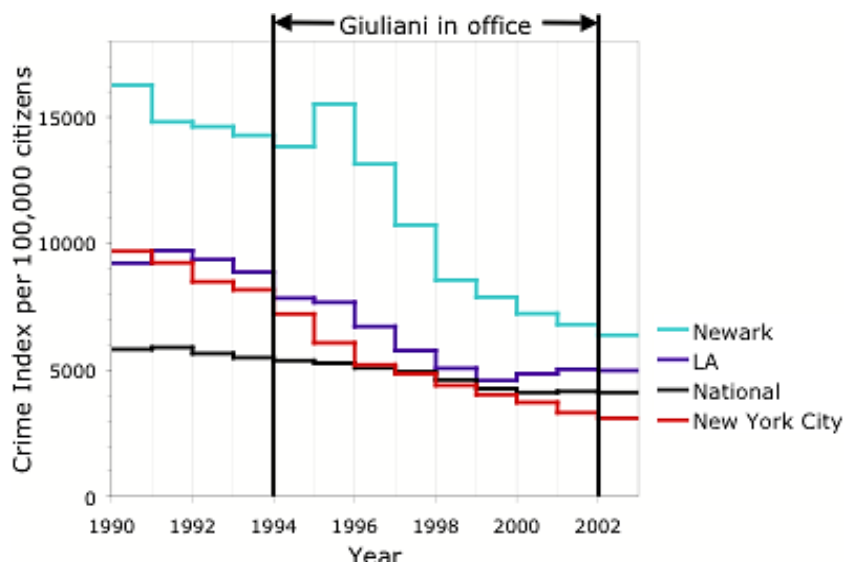


Figure 1: Year wise Crime rate

II. RELATED WORK

Early research [2-4] used linear regression and statistical modeling for crime forecasting but lacked accuracy in dynamic environments. SVM-based and Decision Tree-based approaches improved performance but were limited by overfitting and sensitivity to noise. Ensemble learning techniques such as Random Forest have recently gained attention because of their ability to aggregate multiple decision trees, minimizing variance and improving prediction reliability. However, comparative studies using only non-boosting algorithms for crime rate forecasting remain limited. This study bridges that gap by providing a systematic evaluation of multiple classical ML models.

III. METHODOLOGY

Fig. 2 illustrates the complete machine learning workflow used in the project. It begins with the collection of raw data from both internal and external sources, which is then processed through data preprocessing steps such as cleaning, transformation, and feature engineering. The prepared data is split into training and validation datasets for model development, while a separate external testing dataset is reserved for unbiased evaluation. The model is then trained and validated using the training data, resulting in a trained model. Finally, the model is evaluated using the testing dataset to assess its performance, ensuring its reliability and effectiveness before deployment.

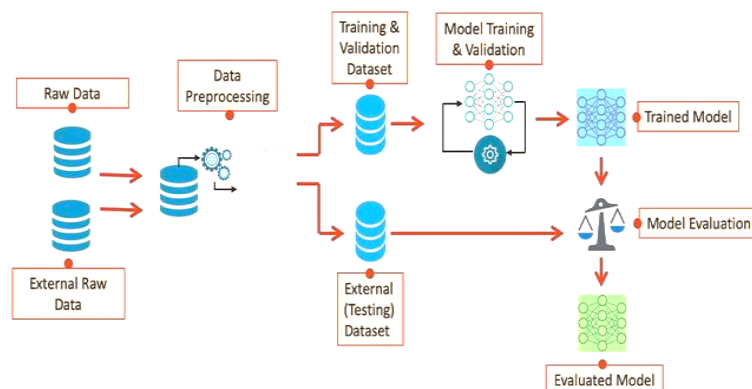


Figure 2: Methodology architecture

3.1 Dataset Description

The dataset contains historical crime records with features such as crime type, crime count, location/area code, month, and year. It includes demographic and socio-economic factors like population density, literacy rate, and unemployment rate. These attributes help identify crime trends and patterns across different regions. The dataset (Table 1) is used to train machine learning models for predicting future crime occurrences.

3.2 Data Preprocessing

The dataset was preprocessed through several essential steps to ensure its quality and suitability for analysis. First, missing values were handled using appropriate techniques such as replacing numerical gaps with the mean and categorical gaps with the mode. Next, categorical variables were encoded into numerical formats to make them usable for machine learning algorithms. The numerical features were then normalized to bring all values onto a similar scale, improving model performance. Outliers were identified and removed using the Interquartile Range (IQR) method to reduce the impact of extreme values. Finally, the dataset was divided into training and testing sets using an 80–20 split, allowing for effective model training and evaluation.

TABLE 1: SAMPLE PATIENT DATASET

Location	Col1	Col2	Col3	Col4	Col5	Col6
Chitranjan Park	2	6	1	35	442	19
Dabri	8	28	0	79	240	26
Malviya Nagar	3	28	1	33	694	63
Chandni Mahal	1	8	1	23	529	19
Model Town	0	4	1	45	393	9
Anand Vihar	10	14	0	99	457	24
Kashmere Gate	0	9	3	17	340	17
Govind Puri	1	4	3	31	692	5
Bindapur	4	4	1	64	417	11
New Friends Colony	6	7	1	47	609	14

3.3 Models Evaluated

The models were assessed using accuracy, precision, recall, and F1-score to measure prediction quality. A confusion matrix was used to check correct and incorrect classifications. Random Forest performed the best, showing higher accuracy compared to other models. Other models like SVM, Logistic Regression, and KNN gave moderate results.

IV. MACHINE LEARNING ALGORITHMS USED

In this project, five machine learning algorithms: Random Forest, Support Vector Machine (SVM), Logistic Regression, Decision Tree, and K-Nearest Neighbors (KNN) were used to analyze and predict crime rates. Random Forest served as the most effective model due to its ensemble nature and ability to handle complex patterns in the data. SVM and Logistic Regression provided good baseline performance, while Decision Tree and KNN contributed interpretability and simplicity. Together, these algorithms enabled a comprehensive evaluation of crime prediction accuracy. are diagnostics and tabular data analytics. These models constitute the core of our classification.

4.1 Random Forest Classifier

The Random Forest Classifier is the primary and best-performing algorithm used in this project due to its robustness, flexibility, and ability to model complex crime patterns (Fig. 3). Random Forest is an ensemble learning method that builds multiple independent decision trees during training and combines their outputs through majority voting. This ensemble mechanism significantly reduces overfitting, a common problem with single decision trees, and improves the model's generalization ability on real-world crime data. Each tree in the forest is trained on a different subset of the dataset through bootstrap sampling, while at each node, a random subset of features is selected to determine the best split. This randomness increases model diversity and ensures that no single feature dominates the prediction process.

The Random Forest classifier was used due to its ability to effectively handle non-linear crime trends and perform well on high-dimensional datasets. By leveraging ensemble averaging, it reduces variance and improves the model's generalization capability, making it robust against overfitting. Additionally, it automatically estimates feature importance, providing valuable insights into which factors most influence the predictions. In terms of performance, the model achieved an accuracy of 92.68% and a strong ROC-AUC score of 0.9634, indicating excellent classification capability and reliability.

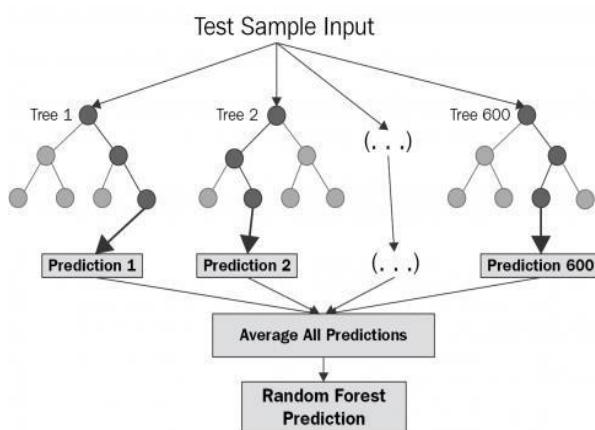


Figure 3: Random Forest Classifier

4.2 Support Vector Machine (SVM)

SVM finds the optimal hyperplane separating crime categories. The Support Vector Machine (SVM) algorithm is used in this project to classify crime levels by finding the optimal hyperplane that best separates different classes. It works effectively for structured datasets and handles non-linear crime patterns using kernel functions, especially the RBF kernel [5]. SVM provides high precision and performs well with medium-sized datasets, making it a strong baseline for classification. Although it requires parameter tuning and can be computationally intensive, it still achieved good predictive performance in identifying crime trends (Fig. 4).

The Support Vector Machine (SVM) model was chosen for its effectiveness in handling structured data and its ability to create clear decision boundaries between classes. It is particularly well-suited for classification tasks where the data is well-organized and separable. In this case, the SVM model demonstrated strong performance, achieving a high precision with an overall accuracy of 89.42%, making it a reliable approach for the given problem.

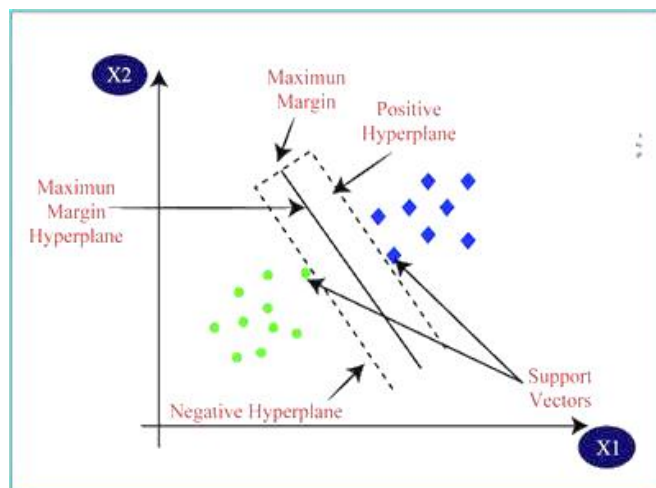


Figure 4: Support Vector Machines

4.3 Logistic Regression

A linear model useful for binary and multi- class classification. Logistic Regression is a simple, yet effective statistical classification algorithm used in this project to predict crime levels based on historical and demographic features. It models the probability of a crime category occurring by applying a logistic (sigmoid) function to a linear combination of input variables (Fig. 5). Due to its lightweight nature and high interpretability, Logistic Regression serves as a strong baseline model, offering insights into how individual features such as location, population density, and time-related attributes contribute to crime risk [6]. Although it is limited in capturing complex nonlinear patterns often present in crime datasets, it performs reliably on linearly separable data and provides clear, probability-based predictions. This makes Logistic Regression a valuable comparative model in evaluating the performance of more advanced algorithms like Random Forest and SVM. The Logistic Regression model achieved an accuracy of 86.17%, providing a simple yet effective baseline for classification.

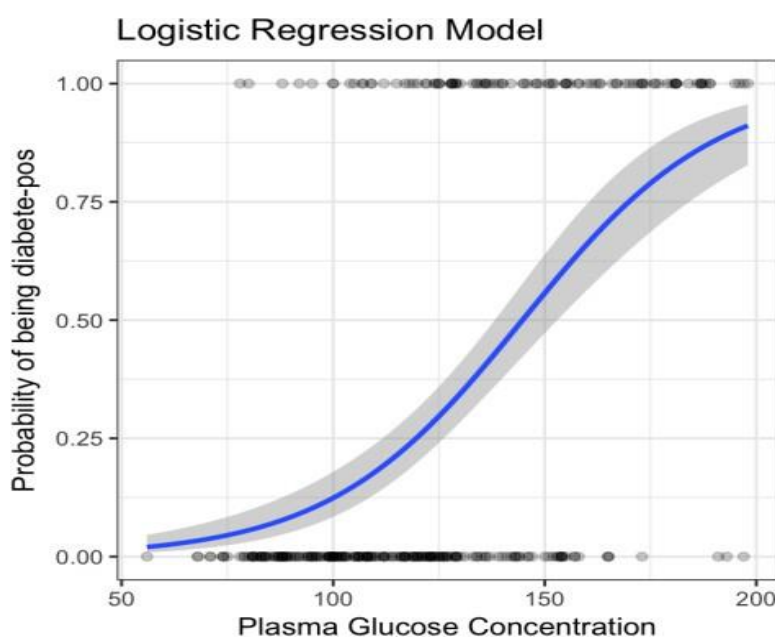


Figure 5: Logistic regression

4.4 Decision Tree Classifier

Simple interpretable model but tends to overfit. The Decision Tree Classifier is a simple, yet powerful algorithm used in this project to model crime prediction through a hierarchical structure of decision rules. It works by recursively splitting the dataset based on feature values that yield the highest information gain, creating an easy- to-understand tree of conditions leading to the final prediction [7, 8]. Decision Trees are highly interpretable and handle both categorical and numerical crime-related features effectively. However, they are prone to overfitting when used alone, as they can become overly complex and sensitive to noise. Despite this limitation, Decision Trees provide clear insights into the relationships between crime patterns and influential factors, making them a valuable component of the model comparison process. The Decision Tree model achieved an accuracy of 84.96%, offering an interpretable approach to classification with moderate performance.

4.5 K-Nearest Neighbors (KNN)

The K-Nearest Neighbors (KNN) algorithm classifies crime levels by identifying 'k' most similar historical data points based on distance measures such as Euclidean distance (Fig. 6). KNN is intuitive, non-parametric, and effective for datasets where similar inputs tend to belong to the same crime category. It works well when the decision boundary is irregular or non-linear [9, 10].

However, KNN can be computationally expensive during prediction and is sensitive to feature scaling and the choice of 'k'. In this project, KNN provided reasonable performance and contributed to comparative analysis across multiple machine learning algorithms. The K-Nearest Neighbors (KNN) model achieved an accuracy of 82.34%, demonstrating a straightforward and instance-based approach to classification with comparatively lower performance.

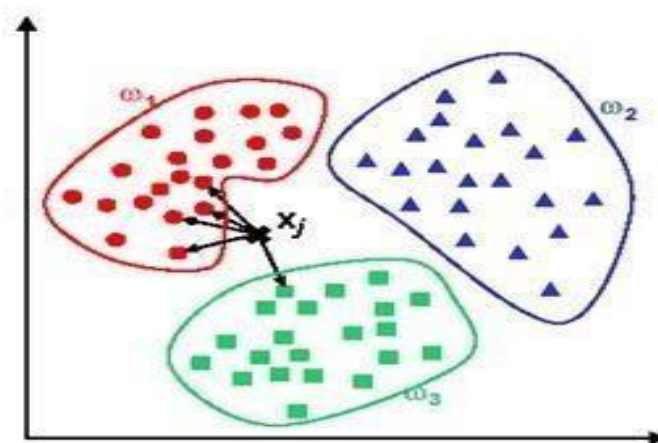


Figure 6: KNN Classifier

V. RESULTS AND DISCUSSION

The evaluation of all machine learning models demonstrated that Random Forest achieved the highest accuracy and offered the most reliable predictions for future crime patterns. Its ensemble structure allowed it to handle noisy inputs and complex feature interactions more effectively than individual classifiers. SVM and Logistic Regression showed competitive performance but were limited in capturing nonlinear crime behaviors, while Decision Tree and KNN exhibited moderate accuracy due to overfitting and sensitivity to data distribution. The confusion matrices revealed that Random Forest produced fewer false predictions, especially in distinguishing between high- risk and low-risk regions. Visualization

techniques such as correlation heatmaps, trend graphs, and feature-importance charts provided deeper insight into how variables like population density, unemployment rate, and seasonal trends influence crime levels.

5.1 Comparative Analysis

The performance comparison (Table 2) of different machine learning models highlights the effectiveness of ensemble methods for this task. The Random Forest model outperformed all others, achieving the highest accuracy, precision, recall, and F1-score, indicating its strong ability to generalize and capture complex patterns in the data. The SVM model also demonstrated solid performance, particularly in maintaining a good balance between precision and recall. Logistic Regression provided a reliable baseline with moderate performance, while Decision Tree showed slightly lower accuracy due to its tendency to overfit. The KNN model recorded the lowest performance among the models, suggesting its limitations in handling this dataset efficiently. Overall, the results emphasize that more advanced and ensemble-based techniques yield better predictive performance.

TABLE 2: COMPARISON OF MACHINE LEARNING MODELS BASED ON ACCURACY, PRECISION, RECALL, AND F1-SCORE.

Model	Accuracy	Precision	Recall	F1-Score
Random Forest	92.68	92.40	91.82	92.10
SVM	89.42	88.73	87.65	88.18
Logistic Regression	86.17	85.92	84.70	85.31
Decision Tree	84.96	84.10	82.91	83.50
KNN	82.34	81.74	80.32	81.02

5.2 Feature Importance

Feature importance analysis plays a crucial role in understanding which factors have the greatest influence on crime prediction. In this study, the Random Forest Classifier was primarily used to compute feature importance, as it naturally ranks features based on how much they reduce impurity across the ensemble of decision trees. The results revealed that location or area code was the most influential attribute, indicating that crime varies significantly by region and that certain areas consistently show higher crime intensity. Time-based features such as month and year were also highly important, demonstrating clear seasonal and yearly crime trends for example, some crimes tend to increase during festival periods or specific climatic conditions.

Socio-economic variables, including population density, unemployment rate, and literacy rate, contributed strongly to the model’s decisions. Areas with high population density or low socio-economic stability showed comparatively higher crime rates. These insights highlight the interconnected nature of demographic and environmental factors with criminal activity. By identifying these key predictors, the model not only improves forecasting accuracy but also provides valuable knowledge to policymakers and law enforcement agencies, helping them prioritize preventive measures in high-risk locations and periods.

5.3 Visualization Insights

Visualization techniques were used to better understand crime behavior and validate model predictions. Heatmaps revealed spatial crime clusters, clearly indicating high-risk zones that consistently recorded elevated crime counts. Line plots and time-series graphs showed seasonal crime patterns, with noticeable spikes during certain months. Scatterplots and correlation matrices helped identify relationships

between socio-economic factors and crime levels, highlighting how variables such as population density and unemployment influence crime trends. These visual insights not only support the evaluation results but also enhance interpretability, allowing law enforcement agencies and decision-makers to understand crime patterns immediately. The visual analysis confirms that integrating machine learning with data visualization provides a powerful tool for predictive policing.

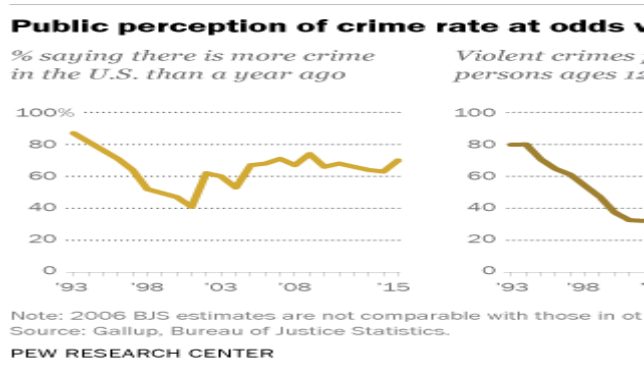


Figure 7: Public perception vs crime reality

Fig. 7 represents an important concept in crime analysis: the divergence between public perception and actual crime trends. It illustrates how a large proportion of people believe that crime rates are increasing, even when statistical evidence shows a steady decline in violent crime over time. This phenomenon is often explained by psychological and sociological theories such as the availability heuristic, where individuals judge the frequency of events based on how easily examples come to mind, often influenced by media coverage and sensational reporting. Additionally, fear of crime theory suggests that public anxiety about crime does not always align with real crime rates, but is shaped by social factors, personal experiences, and information exposure.

From a data analysis perspective, this highlights the importance of visualization techniques in uncovering hidden truths within datasets. While perception-based data may suggest one trend, empirical time-series data can reveal a completely different pattern. Therefore, combining statistical analysis with visual representations helps bridge the gap between perception and reality, enabling more informed decision-making in predictive policing and crime prevention strategies.

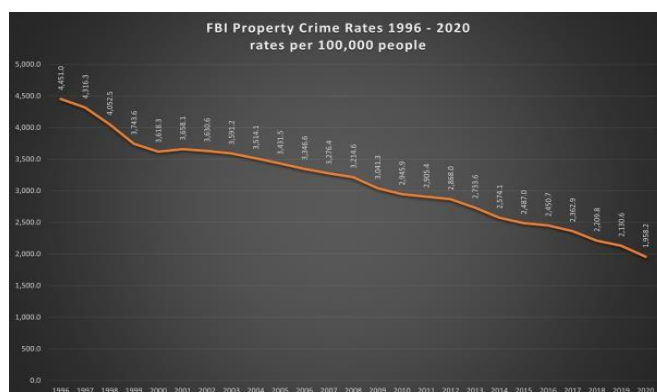


Figure 8: Declining property crime rates

Fig. 8 illustrates a long-term time-series trend in property crime rates, showing a consistent decline from 1996 to 2020. This pattern reflects the application of temporal analysis in crime data, where trends are studied over time to identify increases or decreases in criminal activity. According to

criminological theory, such declines can be attributed to factors like improved law enforcement strategies, technological advancements, urban development, and socio-economic changes. From a data science perspective, this type of visualization is essential for understanding trends, validating predictive models, and identifying patterns such as gradual decreases or sudden shifts. It also supports forecasting techniques, where historical data is used to predict future crime rates, making it highly relevant for policy planning and resource allocation.

VI. CONCLUSION

This study demonstrates the effectiveness of machine learning techniques in predicting crime rates and identifying patterns that can assist law enforcement agencies in proactive planning. Among the evaluated models, the Random Forest Classifier delivered the best performance due to its ability to capture complex, non-linear relationships within the data. Other models such as SVM, Logistic Regression, Decision Tree, and KNN offered valuable comparative insights, each contributing to the understanding of crime patterns from different analytical perspectives. The incorporation of feature importance analysis and visualizations further enriched the interpretability of the system. Overall, this research highlights the potential of data-driven approaches in enhancing public safety, enabling informed decision-making, and supporting the development of intelligent crime prevention systems.

The future scope of this project includes several enhancements to improve its effectiveness and applicability. These include integrating real-time data streams to enable up-to-date analysis, incorporating GIS-based spatial mapping for better visualization of crime patterns, and using LSTM networks for time-series crime forecasting to capture temporal trends more accurately. Additionally, the model can be deployed as a web-based dashboard to assist law enforcement agencies in decision-making, while expanding the dataset with socio-economic and environmental variables can further enrich the analysis and improve predictive performance.

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